AdminLens — Generative AI Module: Developer Guide

Version: 1.0 · **Date:** 10 October 2026 · **Live site:** https://app-list-dev-35635.web.app/generative-ai/

1. About this guide

This guide explains the **Generative AI** part of AdminLens.

You will learn:

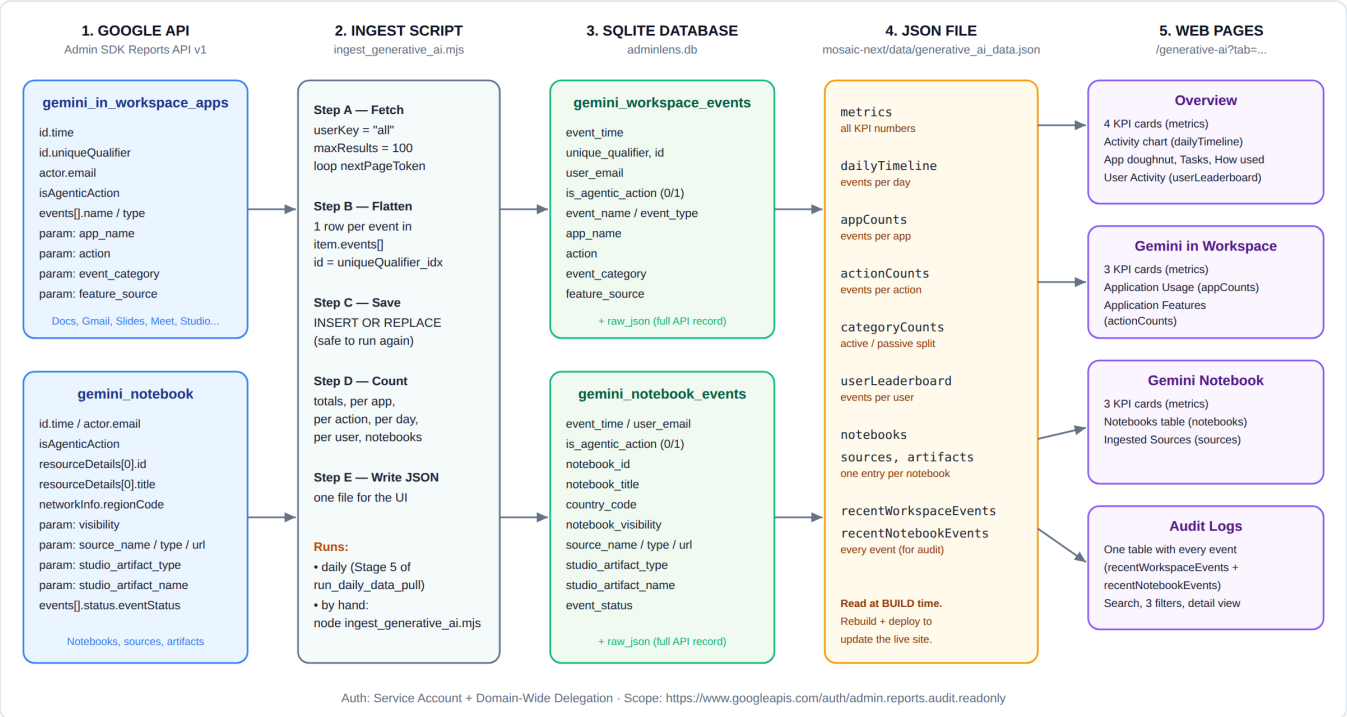
1. Where the data comes from (Google API).
2. How the data moves from Google to the screen.
3. What each page shows, and which data each part uses.
4. How to update the data and publish the website.

> **How to read this guide:** Each section is short. Words in `code style` are exact names from the code. You can search for them in the project.

2. Words used in this guide

Word	Meaning
Event	One action by one user. Example: a user asks Gemini a question.
Workspace event	An event from Gemini inside Google apps (Gmail, Docs, Slides, Meet, Workspace Studio, and others).
Notebook event	An event from Gemini Notebook (also called NotebookLM). Example: a user creates a notebook.
Agentic	Google marked the event as done by an automated agent, not by a person. We read this from the API field `isAgenticAction`.
User-initiated	The opposite of agentic. A person started the action.
Active	The user asked for something (a chat or generated content).
Inactive / Passive	Gemini worked in the background without a request. Example: Meet studio sound.
Ingest	Download data from Google and save it in our database.
KPI card	A small box at the top of a page that shows one important number.

3. How the data moves (API mapping diagram)



The data moves in **5 steps**, from left to right:

1. **Google API.** Google stores audit logs. We read two log streams: `gemini_in_workspace_apps` and `gemini_notebook`.
2. **Ingest script.** The file `ingest_generative_ai.mjs` downloads all events from both streams.
3. **SQLite database.** The script saves each event as one row in `adminlens.db`. There are two tables, one for each stream.
4. **JSON file.** The script counts the events and writes one file: `mosaic-next/data/generative_ai_data.json`.
5. **Web pages.** The Next.js app reads the JSON file and shows the four pages.

> **Important:** The website reads the JSON file **when you build the website**, not when a user opens the page. After new data is ingested, you must **build and deploy again**. If you do not, the live website still shows the old data.

4. The Google API

4.1 Authentication

Item	Value
API	Google Admin SDK **Reports API**, version `reports_v1`
Account type	Google Cloud **Service Account** with **Domain-Wide Delegation**
Key file	`gam-project-83tkn-bccbdf540703.json` (in the project root — keep it secret, do not commit it)
Acts as user	A Workspace super admin (in the code: `subject`)
OAuth scope	`https://www.googleapis.com/auth/admin.reports.audit.readonly`

4.2 The API call

```
GET https://admin.googleapis.com/admin/reports/v1/activity/users/all/applications/{applicationName}
```

Parameter	Value	Why
`userKey`	`all`	Get events for every user in the domain.
`applicationName`	`gemini_in_workspace_apps` or `gemini_notebook`	We call the API **two times**, once for each stream.
`maxResults`	`100`	Google sends 100 records per page.
`pageToken`	value of `nextPageToken`	Repeat the call until `nextPageToken` is empty. Then you have all records.

4.3 How one API record becomes database rows

One API record (an "activity") can contain **more than one event** in its `events[]` list. The script saves **one database row for each event**. The row ID is `uniqueQualifier` + `_` + the event's position in the list (for example `abc123_0`, `abc123_1`).

The script uses `INSERT OR REPLACE`. This means you can run the script many times. It does not create duplicate rows.

4.4 Field mapping — Workspace stream

API stream `gemini_in_workspace_apps` → table `gemini_workspace_events`

Google API field	Database column	Example value
`id.time`	`event_time`	`2026-10-09T13:18:00Z`
`id.uniqueQualifier`	`unique_qualifier` (and part of `id`)	
`id.customerId`	`customer_id`	
`actor.email`	`user_email`	`mickey@gafe.co.za`
`actor.callerType`	`caller_type`	
`isAgenticAction`	`is_agentic_action` (0 or 1)	`1`
`events[].name` / `events[].type`	`event_name` / `event_type`	
parameter `app_name`	`app_name`	`gmail`, `workflows`, `gemini_app`
parameter `action`	`action`	`summarize_unreads`
parameter `event_category`	`event_category`	`active_generate`, `inactive`
parameter `feature_source`	`feature_source`	

Google API field	Database column	Example value
(whole record)	`raw_json`	Full API record, for debugging

4.5 Field mapping — Notebook stream

API stream `gemini_notebook` → table `gemini_notebook_events`

Google API field	Database column	Example value
`id.time`, `actor.email`	`event_time`, `user_email`	
`isAgenticAction`	`is_agentic_action` (0 or 1)	`0`
`resourceDetails[0].id`	`notebook_id`	
`resourceDetails[0].title`	`notebook_title`	`EGD Notebook`
`networkInfo.regionCode`	`country_code`	Two-letter country code
`events[].status.eventStatus`	`event_status`	
parameter `visibility`	`notebook_visibility`	`private`
parameter `source_name` / `source_type` / `source_url`	`source_name` / `source_type` / `source_url`	`URL`, `LOCAL_UPLOAD`
parameter `studio_artifact_type` / `studio_artifact_name`	`studio_artifact_type` / `studio_artifact_name`	Shown on the page as "Audio Overview", "Mind Map"
(whole record)	`raw_json`	Full API record

The full table definitions are in `schema.sql`.

5. The JSON file

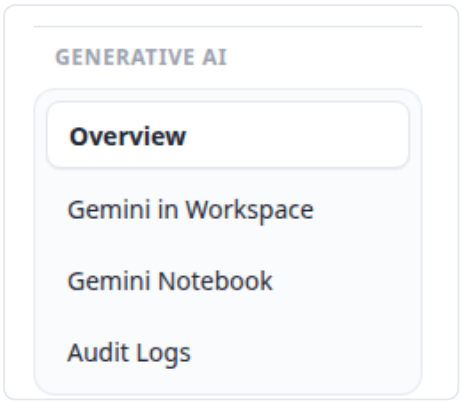
The ingest script writes `mosaic-next/data/generative_ai_data.json`. This table shows each top-level key and where the website uses it.

JSON key	What it contains	Used on page
`updatedAt`	Time the file was created	—
`metrics`	All totals (events, users, agentic count, notebooks, sources, artifacts)	KPI cards on all pages
`dailyTimeline`	Events per day, split by Workspace / Notebook and User / Agentic	Overview — activity chart
`appCounts`	Number of events for each app	Overview — doughnut chart; Gemini in Workspace — Application Usage
`actionCounts`	Number of events for each action	Overview — Capabilities chart; Gemini in Workspace — Application Features
`categoryCounts`	Events per category (`active_conversations` , `active_generate` , `inactive`)	Overview — How AI Is Used
`userLeaderboard`	One entry per user, with event counts and last active date	Overview — User Activity table
`notebooks`	One entry per notebook	Gemini Notebook — Notebooks table
`sources`	One entry per source added to a notebook	Gemini Notebook — Ingested Sources table
`artifacts`	One entry per generated artifact	Gemini Notebook — KPI count (the table uses `notebooks[].artifacts`)
`recentWorkspaceEvents`	Every Workspace event	Audit Logs
`recentNotebookEvents`	Every Notebook event	Audit Logs

6. The pages

The module has **four pages**. They all use the same URL with a different `tab` value:

Menu item	URL
Overview	`/generative-ai?tab=overview`
Gemini in Workspace	`/generative-ai?tab=apps`
Gemini Notebook	`/generative-ai?tab=notebooks`
Audit Logs	`/generative-ai?tab=audit`



All four pages are in one component: `mosaic-next/app/(default)/generative-ai/generative-ai-client.tsx`. The charts are in `mosaic-next/components/generative-ai/`.

6.1 Overview page (`?tab=overview`)

Generative AI Overview

Usage activity, Gemini Notebook projects, and automated workflows across Google Workspace.

Refresh

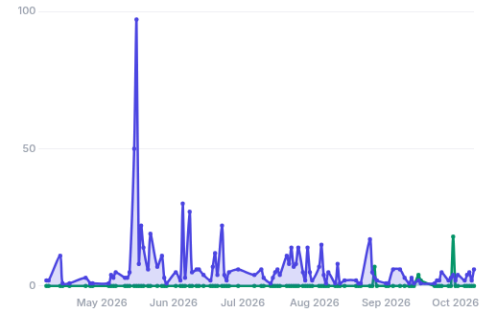
TOTAL ACTIVITY EVENTS	ACTIVE USERS	AGENTIC FLOWS	GEMINI NOTEBOOK PROJECTS
733	30	26	9
702 Workspace · 31 Gemini Notebook	Licensed accounts engaged	Workspace Studio automated tasks	9 Sources · 7 Artifacts

Generative AI Activity

Daily event activity across the domain.

Workspace vs Gemini Notebook | User vs Agentic

WORKSPACE APPS	GEMINI NOTEBOOK	PEAK ACTIVITY DAY
702 (95.8%)	31 (4.2%)	97 events (May 16)



AI Activity by Application

Generative AI events across connected Workspace tools.



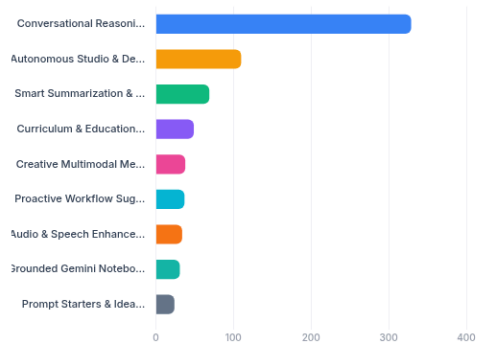
Gemini 294 · Workspace Studio 124 · Google Slides 96 · Gmail 53 · Classroom 49 · Google Meet 31 · Gemini Notebook 31 · Google Vids 29 · Google Docs 24 · Google Forms 2

AI Capabilities & Tasks

What tasks users and agents are delegating to Generative AI.

Core Pillars | Top Actions

Top Driver: Conversational Reasoning & Chat (45.6%) · **Autonomous Automation:** 110 Studio Actions (15.3%)

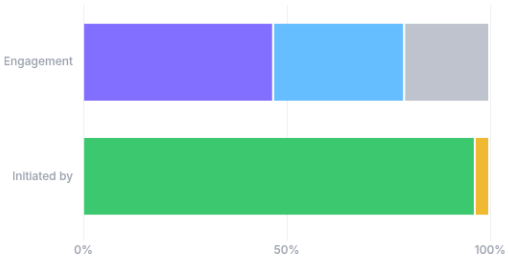


How AI Is Used

ACTIVELY DRIVEN BY USERS

79% 555 events

47% Conversations · **32%** Content generation · **21%** Passive / background · **96%** People · **4%** Agents



Passive features run without a deliberate prompt — high passive share can mean AI is switched on but not yet adopted.

User Activity

Search user email...

USER	WORKSPACE EVENTS	NOTEBOOK EVENTS	TOTAL EVENTS	LAST ACTIVE
plutt@gafe.co.za	139	0	139	Jun 19, 2026
themba@gafe.co.za	129	7	136	Sep 11, 2026
mickey@gafe.co.za	100	0	100	Oct 9, 2026
alister@gafe.co.za	49	20	69	Sep 30, 2026
karien@gafe.co.za	34	4	38	Sep 30, 2026
farrakhan@gafe.co.za	36	0	36	Aug 5, 2026
darklighter@gafe.co.za	36	0	36	May 16, 2026
kevin@gafe.co.za	31	0	31	Sep 7, 2026
deidre@gafe.co.za	27	0	27	Oct 9, 2026
mazk@gafe.co.za	16	0	16	May 19, 2026

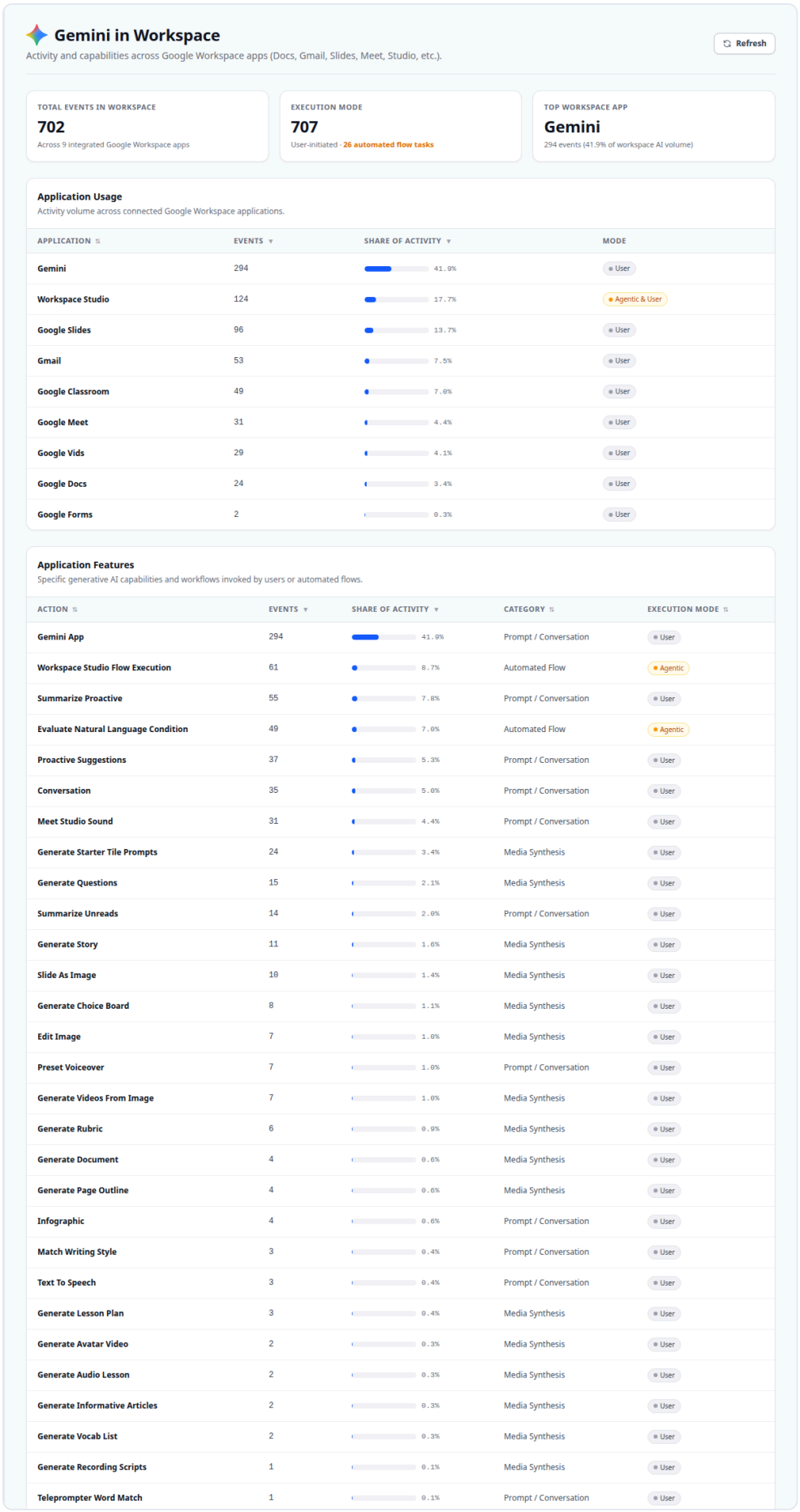
Showing 1–10 of 30 accounts

10 per page · ← Prev · Page 1 / 3 · Next →

Part of the page	What it shows	Data source
KPI: Total Activity Events	All events (Workspace + Notebook)	`metrics.totalEvents`
KPI: Active Users	Users with at least one event	`metrics.totalActiveUsers`
KPI: Agentic Flows	Events where Google set `isAgenticAction = true`	`metrics.agenticEventsCount`
KPI: Gemini Notebook Projects	Number of notebooks, sources, artifacts	`metrics.notebooksCount` and others
Chart: Generative AI Activity	Line chart, events per day. Two buttons switch the view: *Workspace vs Gemini Notebook* or *User vs Agentic*.	`dailyTimeline` · `gen-ai-timeline-chart.tsx`
Chart: AI Activity by Application	Doughnut chart. Share of events for each app. Click a label to hide or show that app.	`appCounts` · `gen-ai-apps-doughnut.tsx`
Chart: AI Capabilities & Tasks	Bar chart. *Core Pillars* groups actions into 9 groups. *Top Actions* shows the 8 biggest actions. The two labels at the top are calculated from the data.	`actionCounts` · `gen-ai-capabilities-bar.tsx`
Chart: How AI Is Used	Two bars. Top bar: Conversations / Content generation / Passive. Bottom bar: People / Agents.	`categoryCounts` · `gen-ai-engagement-mix.tsx`
Table: User Activity	One row per user. Search by email. Click a column title to sort. 10, 25, 50 or 100 rows per page.	`userLeaderboard`

The four charts are in a 2 × 2 grid. Cards in the same row always have the same height. The charts cannot be closed or hidden.

6.2 Gemini in Workspace page (`?tab=apps`)



Gemini in Workspace

Activity and capabilities across Google Workspace apps (Docs, Gmail, Slides, Meet, Studio, etc.).

Refresh

TOTAL EVENTS IN WORKSPACE	EXECUTION MODE	TOP WORKSPACE APP
702	707	Gemini
Across 9 integrated Google Workspace apps	User-initiated · 26 automated flow tasks	294 events (41.9% of workspace AI volume)

Application Usage

Activity volume across connected Google Workspace applications.

APPLICATION	EVENTS	SHARE OF ACTIVITY	MODE
Gemini	294	41.9%	User
Workspace Studio	124	17.7%	Agentic & User
Google Slides	96	13.7%	User
Gmail	53	7.5%	User
Google Classroom	49	7.0%	User
Google Meet	31	4.4%	User
Google Vids	29	4.1%	User
Google Docs	24	3.4%	User
Google Forms	2	0.3%	User

Application Features

Specific generative AI capabilities and workflows invoked by users or automated flows.

ACTION	EVENTS	SHARE OF ACTIVITY	CATEGORY	EXECUTION MODE
Gemini App	294	41.9%	Prompt / Conversation	User
Workspace Studio Flow Execution	61	8.7%	Automated Flow	Agentic
Summarize Proactive	55	7.8%	Prompt / Conversation	User
Evaluate Natural Language Condition	49	7.0%	Automated Flow	Agentic
Proactive Suggestions	37	5.3%	Prompt / Conversation	User
Conversation	35	5.0%	Prompt / Conversation	User
Meet Studio Sound	31	4.4%	Prompt / Conversation	User
Generate Starter Tile Prompts	24	3.4%	Media Synthesis	User
Generate Questions	15	2.1%	Media Synthesis	User
Summarize Unreads	14	2.0%	Prompt / Conversation	User
Generate Story	11	1.6%	Media Synthesis	User
Slide As Image	10	1.4%	Media Synthesis	User
Generate Choice Board	8	1.1%	Media Synthesis	User
Edit Image	7	1.0%	Media Synthesis	User
Preset Voiceover	7	1.0%	Prompt / Conversation	User
Generate Videos From Image	7	1.0%	Media Synthesis	User
Generate Rubric	6	0.9%	Media Synthesis	User
Generate Document	4	0.6%	Media Synthesis	User
Generate Page Outline	4	0.6%	Media Synthesis	User
Infographic	4	0.6%	Prompt / Conversation	User
Match Writing Style	3	0.4%	Prompt / Conversation	User
Text To Speech	3	0.4%	Prompt / Conversation	User
Generate Lesson Plan	3	0.4%	Media Synthesis	User
Generate Avatar Video	2	0.3%	Media Synthesis	User
Generate Audio Lesson	2	0.3%	Media Synthesis	User
Generate Informative Articles	2	0.3%	Media Synthesis	User
Generate Vocab List	2	0.3%	Media Synthesis	User
Generate Recording Scripts	1	0.1%	Media Synthesis	User
Teleprompter Word Match	1	0.1%	Prompt / Conversation	User

Part of the page	What it shows	Data source
KPI: Total Events in Workspace	Events from Google apps only (no Notebook events)	`metrics.totalWorkspaceEvents`
KPI: Execution Mode	User-initiated events and agentic events	`metrics.userInitiatedCount`, `metrics.agenticEventsCount`
KPI: Top Workspace App	The app with the most events	`appCounts` (highest value)
Table: Application Usage	One row per app: events, share bar, mode. Click a column title to sort.	`appCounts`
Table: Application Features	One row per action: events, share bar, category, execution mode. Click a column title to sort.	`actionCounts`

6.3 Gemini Notebook page (`?tab=notebooks`)

Gemini Notebook

Active notebook projects, ingested reference sources, and generated studio artifacts.

Refresh

TOTAL NOTEBOOKS	INGESTED SOURCES	GENERATED STUDIO ARTIFACTS
9	9	7
2 shared internally · 7 private	Reference documents, PDFs & web URLs	Audio overviews, mind maps & study guides

Notebooks

Active notebooks created and managed in the domain.

NOTEBOOK TITLE	OWNER	VISIBILITY	SOURCES	ARTIFACTS	EVENTS	LAST ACTIVE
Analysis of the 2019 Ransomware Attack at Maastricht University	alister@gafe.co.za	Private	8	Audio Overview, Mind Map	14	Sep 30, 2026
[Training] Senior Phase Coding and Robotics Curriculum	karien@gafe.co.za	Shared Internally	0	Audio Overview	2	Sep 30, 2026
[Training] The Boy in the Striped Pajamas	karien@gafe.co.za	Shared Internally	0	Mind Map	2	Sep 30, 2026
Untitled notebook	alister@gafe.co.za	Private	0	—	1	Sep 16, 2026
Gemini for Teachers and Google Apps Manager Guide	alister@gafe.co.za	Private	0	Mind Map	4	Sep 16, 2026
Untitled notebook	alister@gafe.co.za	Private	0	—	1	Sep 15, 2026
EGD Notebook	themba@gafe.co.za	Private	1	Mind Map, Mind Map	5	Aug 27, 2026
Untitled notebook	themba@gafe.co.za	Private	0	—	1	Aug 27, 2026
Untitled notebook	themba@gafe.co.za	Private	0	—	1	Aug 27, 2026

Showing 1–9 of 9 notebooks — 10 per page — ← Prev — Page 1 / 1 — Next →

Ingested Sources

External documents, URLs, and uploaded files attached to Gemini Notebook notebooks.

DOCUMENT / SOURCE NAME	TYPE	TARGET NOTEBOOK	USER	ADDED DATE
Cybersecurity Challenges in Higher Education - AICUP https://aicup.org/wp-content/uploads/2023/11/Cybersecurity-IHE.pdf	URL	Analysis of the 2019 Ransomware Attack at Maastricht University	alister@gafe.co.za	Sep 30, 2026
Cyber security breaches survey 2025/2026: education institutions findings - GOV.UK https://www.gov.uk/government/statistics/cyber-security-breaches-survey-20252026/cyber-securi...	URL	Analysis of the 2019 Ransomware Attack at Maastricht University	alister@gafe.co.za	Sep 30, 2026
ANALYSIS OF THE 2019 RANSOMWARE ATTACK AT THE MAASTRICHT UNIVERSITY - Centri — Manage https://centri.unibo.it/computational-social-science/en/our-work/our-students-papers/antonini-gi...	URL	Analysis of the 2019 Ransomware Attack at Maastricht University	alister@gafe.co.za	Sep 30, 2026
Zero Trust Architecture - NIST Technical Series Publications https://nvlpubs.nist.gov/nistpubs/specialpublications/NIST.SP.800-207.pdf	URL	Analysis of the 2019 Ransomware Attack at Maastricht University	alister@gafe.co.za	Sep 30, 2026
NIST SP 800-171 Compliance Guide for Colleges & Universities - UpGuard https://www.upguard.com/blog/nist-sp-800-171-compliance-guide-for-colleges-universities	URL	Analysis of the 2019 Ransomware Attack at Maastricht University	alister@gafe.co.za	Sep 30, 2026
Multi-Factor Authentication: Why It Matters for Higher Education and Research - IU Blogs https://blogs.iu.edu/renisac/2025/08/02/multi-factor-authentication-why-it-matters-for-higher-ed...	URL	Analysis of the 2019 Ransomware Attack at Maastricht University	alister@gafe.co.za	Sep 30, 2026
Data Breach Response Checklist - | Protecting Student Privacy https://studentprivacy.ed.gov/sites/default/files/resource_document/file/checklist_data_breach_r...	URL	Analysis of the 2019 Ransomware Attack at Maastricht University	alister@gafe.co.za	Sep 30, 2026
Cost of a Data Breach Report 2025 The AI Oversight Gap - Baker Donelson https://www.bakerdonelson.com/webfiles/Publications/20250822_Cost-of-a-Data-Breach-Report-...	URL	Analysis of the 2019 Ransomware Attack at Maastricht University	alister@gafe.co.za	Sep 30, 2026
CAPS FET _ ENGINEERING GRAPICHS & DESIGN _ GR 10-12 _ Web_8899.pdf	LOCAL_UPLOAD	EGD Notebook	themba@gafe.co.za	Aug 27, 2026

Showing 1–9 of 9 sources — 10 per page — ← Prev — Page 1 / 1 — Next →

Part of the page	What it shows	Data source
KPI: Total Notebooks	Number of notebooks, and how many are shared or private	`metrics.notebooksCount`, `notebooks[].visibility`
KPI: Ingested Sources	Documents, PDFs and web links added to notebooks	`metrics.sourcesIngestedCount`
KPI: Generated Studio Artifacts	Audio overviews, mind maps, study guides	`metrics.artifactsCount`
Table: Notebooks	Title, owner, visibility, number of sources, artifacts, events, last active date	`notebooks`
Table: Ingested Sources	Source name and link, type (`URL`, `LOCAL_UPLOAD`), notebook, user, date added	`sources`

6.4 Audit Logs page (`?tab=audit`)

Generative AI Audit Logs

Granular activity events, user prompts, and automated agentic executions.

Refresh

Search user, action, notebook, source... | All Applications | All Status | All Actor Modes

TIMESTAMP	USER	APPLICATION	ACTION	TARGET / CONTEXT	ENGAGEMENT	MODE	DETAILS
Oct 9 13:18	mickey@gafe.co.za	Google Meet	meet studio sound	—	Inactive	User	View
Oct 9 13:05	mickey@gafe.co.za	Workspace Studio	evaluate natural language condition	Google Calendar for Workspace Studio	Active	User	View
Oct 9 13:05	mickey@gafe.co.za	Workspace Studio	workspace studio flow execution	Google Calendar for Workspace Studio	Active	User	View
Oct 9 13:04	mickey@gafe.co.za	Google Meet	meet studio sound	—	Inactive	User	View
Oct 9 07:17	deidre@gafe.co.za	Workspace Studio	summarize unreads	Get a daily summary of unread emails	Active	Agentic	View
Oct 9 07:17	deidre@gafe.co.za	Workspace Studio	workspace studio flow execution	Get a daily summary of unread emails	Active	Agentic	View
Oct 8 07:17	deidre@gafe.co.za	Workspace Studio	summarize unreads	Get a daily summary of unread emails	Active	Agentic	View
Oct 8 07:17	deidre@gafe.co.za	Workspace Studio	workspace studio flow execution	Get a daily summary of unread emails	Active	Agentic	View
Oct 7 13:50	mickey@gafe.co.za	Workspace Studio	evaluate natural language condition	Google Calendar for Workspace Studio	Active	User	View
Oct 7 13:50	mickey@gafe.co.za	Workspace Studio	workspace studio flow execution	Google Calendar for Workspace Studio	Active	User	View
Oct 7 13:48	mickey@gafe.co.za	Google Meet	meet studio sound	—	Inactive	User	View
Oct 7 07:17	deidre@gafe.co.za	Workspace Studio	summarize unreads	Get a daily summary of unread emails	Active	Agentic	View
Oct 7 07:17	deidre@gafe.co.za	Workspace Studio	workspace studio flow execution	Get a daily summary of unread emails	Active	Agentic	View
Oct 6 15:35	mickey@gafe.co.za	Workspace Studio	evaluate natural language condition	Google Calendar for Workspace Studio	Active	User	View
Oct 6 15:35	mickey@gafe.co.za	Workspace Studio	workspace studio flow execution	Google Calendar for Workspace Studio	Active	User	View
Oct 6 07:17	deidre@gafe.co.za	Workspace Studio	summarize unreads	Get a daily summary of unread emails	Active	Agentic	View
Oct 6 07:17	deidre@gafe.co.za	Workspace Studio	workspace studio flow execution	Get a daily summary of unread emails	Active	Agentic	View
Oct 5 07:17	deidre@gafe.co.za	Workspace Studio	summarize unreads	Get a daily summary of unread emails	Active	Agentic	View
Oct 5 07:17	deidre@gafe.co.za	Workspace Studio	workspace studio flow execution	Get a daily summary of unread emails	Active	Agentic	View
Oct 2 13:20	mickey@gafe.co.za	Workspace Studio	evaluate natural language condition	Google Calendar for Workspace Studio	Active	User	View

Part of the page	What it shows	Data source
Search box	Searches user, action, app, notebook title and source name	—
Filter: Applications	Show events from one app only	`appName`
Filter: Status	*Active Usage* or *Inactive / Ambient*	`category` (`inactive` = inactive)
Filter: Actor Modes	*User Initiated* or *Agentic Automated*	`isAgenticAction`
Table	Time, user, app, action, target, engagement, mode. Newest events first.	`recentWorkspaceEvents` + `recentNotebookEvents`
View (Event Details)	A pop-up window with the event fields and the full event as JSON	The selected event from the JSON file

7. How to update data and publish

7.1 Automatic update (every day)

The daemon `run_sync_daemon.mjs` starts `run_daily_data_pull.mjs` every 24 hours. **Stage 5** of that script runs `ingest_generative_ai.mjs`.

7.2 Manual update

Run these commands in order:

```
# 1. Download new events and write the JSON file
cd /home/alister/MCP-servers/adminlens-investigation
node ingest_generative_ai.mjs

# 2. Build the website (this reads the new JSON file)
cd mosaic-next
npm run build

# 3. Publish to Firebase Hosting
npx firebase-tools deploy --only hosting
```

Firebase project: `app-list-dev-35635`. The build output folder is `mosaic-next/out` (see `firebase.json`).

7.3 Check your work before you publish

```
cd mosaic-next
npx tsc --noEmit          # TypeScript check — must show no errors
npx eslint "app/(default)/generative-ai" components/generative-ai
```

8. Known issues and things to be careful with

Please read this list before you change the code. These are not bugs in Google's data. They are decisions or gaps in our code.

1. **The Refresh button works only on a local computer.** It calls `http://localhost:3333/api/generative-ai/sync` (in `server.mjs`). On the live Firebase site, there is no server, so the button cannot work. To update the live site, use section 7.2.
2. **"Agentic" has two different meanings.**
   - The KPI **Agentic Flows (26)** counts events where **Google** set `isAgenticAction = true`.
   - The label **Autonomous Automation (110 Studio Actions)** on the Capabilities chart counts events with a Workspace Studio **action name** (`classic_use_case_workspace_studio_flow_execution` and `evaluate_natural_language_condition`).
   - Google marks some Studio flows as started by a user. So these two numbers are different. This is correct, but users may ask about it.
3. **"Execution Mode" on the Gemini in Workspace page includes Notebook events.** The number (707) is bigger than "Total Events in Workspace" (702). This is because `metrics.userInitiatedCount` counts Workspace **and** Notebook events. To fix it, add a Workspace-only count in the ingest script.
4. **"Owner" of a notebook is a guess.** The API does not send the notebook owner. We use the email of the **first event** we see for that notebook.
5. **Google Forms is missing from the Applications filter** on the Audit Logs page. It has events (2), but there is no option for it in the drop-down list.
6. **The "Agentic & User" mode** in the Application Usage table is fixed for Workspace Studio. It is not calculated from the data.
7. **App names are mapped by hand.** If Google adds a new `app_name`, add a name and colour in `APP_META` in `gen-ai-apps-doughnut.tsx`, and add a filter option on the Audit Logs page.

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9. Important files

File	Purpose
`ingest_generative_ai.mjs`	Downloads events, saves to SQLite, writes the JSON file
`schema.sql`	Database table definitions
`run_daily_data_pull.mjs` / `run_sync_daemon.mjs`	Daily automatic update
`server.mjs`	Local API used by the Refresh button
`mosaic-next/data/generative_ai_data.json`	Data file read by the website
`mosaic-next/app/(default)/generative-ai/page.tsx`	Reads the JSON file at build time
`mosaic-next/app/(default)/generative-ai/generative-ai-client.tsx`	All four pages, tables, filters
`mosaic-next/components/generative-ai/*.tsx`	The four Overview charts and the chart card
`mosaic-next/components/ui/sidebar.tsx`	Sidebar menu
`docs/dev-guide/img/api-mapping-diagram.svg`	The diagram in this guide (you can edit it)